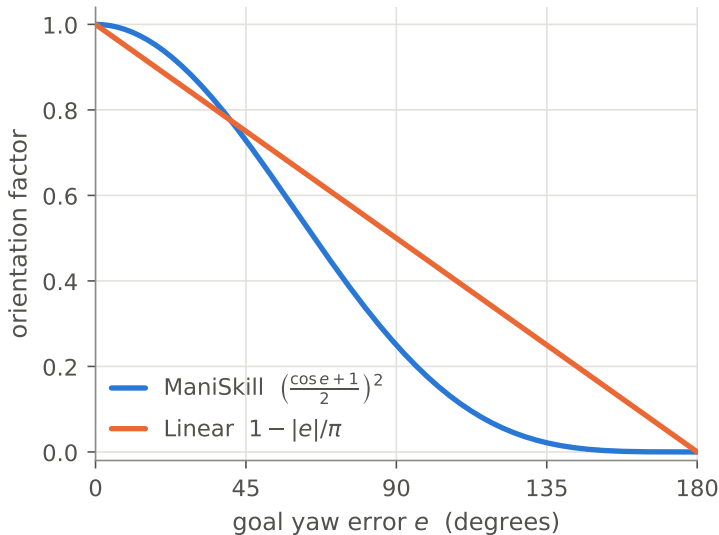


$$\text{ManiSkill: } r = \frac{1}{3} \left[w \left(\frac{\cos e + 1}{2} \right)^2 + (1 - w)(1 - \tanh 5d)^2 + \frac{1}{20} \sqrt{1 - \tanh 5d_{ee}} \right]$$

$$\text{Linear variant: } r = \frac{1}{3} \left[w \left(1 - \frac{|e|}{\pi} \right) + (1 - w)(1 - \tanh 5d)^2 + \frac{1}{20} \sqrt{1 - \tanh 5d_{ee}} \right]$$

$w = 0.5$; d measures the T's centre of mass to the goal, d_{ee} the end-effector to that same point. At goal (overlap ≥ 0.90) the bracket is replaced by 3, so $r = 1$: a value the shaped terms cannot reach.

(a) Orientation factors



(b) Distance factors

